Low-demo adaptation of SmolVLA on LIBERO

A leakage-controlled, compute-limited study of 0/5/10/25 demonstrations

Abstract

We adapt a 450M-parameter SmolVLA policy to the first three LIBERO-Goal tasks. The experiment preserves the requested task set, demonstration budgets, two training seeds and 20 simulator episodes per task/checkpoint. Compute is limited only through 5,000 seen-pretraining updates and 500 target-adaptation updates. We compare ordinary fine-tuning with rank-16 LoRA plus mild image augmentation, include a paired wrong-language control, analyze three real failures, and rank checkpoints without reward. The measured LoRA-minus-baseline differences at 5/10/25 demonstrations are +0.0, -6.7, and +0.0 percentage points. Only the naive 10-demo condition succeeds (8/120); the proposed method records no success at any budget.

1. Experimental design

The seen pool is the available `libero_90` conversion: 73 task ids, 3,921 episodes and 569,249 frames. Targets are simulator task ids 0-2 in `libero_goal`. The simulator and dataset use different task orders, so demonstrations are joined by exact canonical prompt, not numeric task id. This prevents silent training on the wrong target.

Simulator id	Dataset task_index	Instruction	First five episode ids
0	9	open the middle drawer of the cabinet	20, 26, 31, 42, 58
1	7	put the bowl on the stove	13, 15, 16, 22, 36
2	4	put the wine bottle on top of the cabinet	6, 10, 17, 18, 25

Training and evaluation

Component	Fixed specification
Base / seen	lerobot/smolvla_base; 5,000 updates x seeds 17, 29 on all 73 seen tasks
Naive baseline	500 updates; standard action-expert fine-tuning; no transforms
Proposed method	rank-16 LoRA; alpha 16; at most two random mild image transforms
Evaluation	300-step LIBERO horizon; 20 episodes per task/checkpoint; seed 1000
Hardware	RTX 4070 Ti SUPER 16 GB; PyTorch 2.7.1+cu126; LeRobot 9e145849...

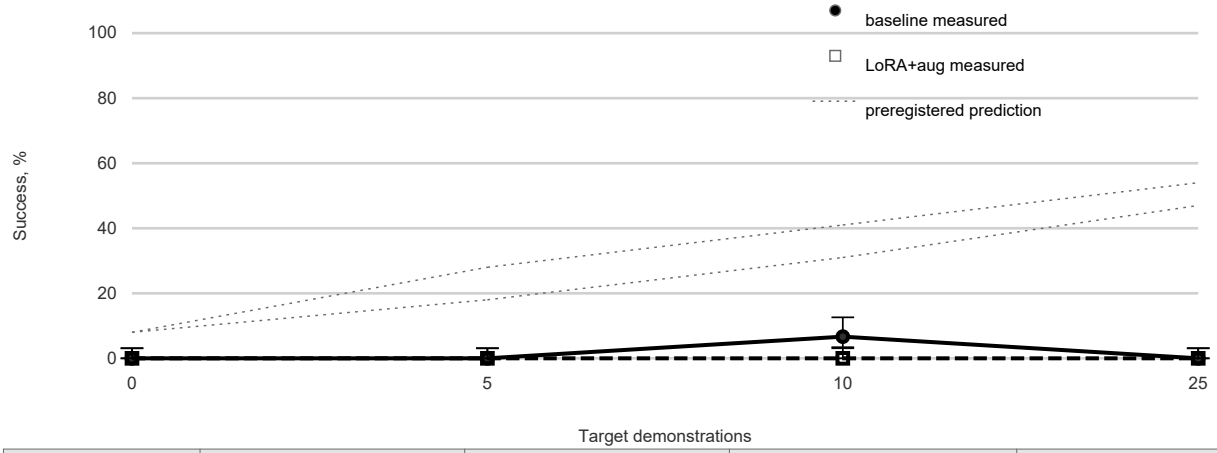
Preregistration

Before adaptation runs we predicted a monotone curve and the largest LoRA advantage at five demonstrations. Prediction SHA-256 and exact nested episode prefixes were fixed before training; manifest digest: 82ddbb8c6248486c444c8ffd... .

The public 40-task LIBERO checkpoint was rejected because its training universe overlaps the held-out goal tasks. All measured target checkpoints descend from our own seen run.

2. Results

Cost curve



Demos	Zero-shot	Naive	LoRA+aug	Difference
0	0.0%	-	-	-
5	-	0.0%	0.0%	+0.0 pp
10	-	6.7%	0.0%	-6.7 pp
25	-	0.0%	0.0%	+0.0 pp

Markers and 95% Wilson intervals are measured simulator outcomes; faint dotted lines are the preregistered expectations. The shared zero-shot point is evaluated from both seen seeds. No incomplete run is imputed.

Language control

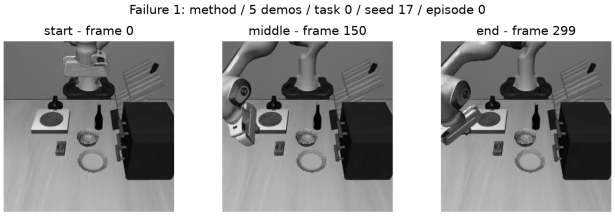
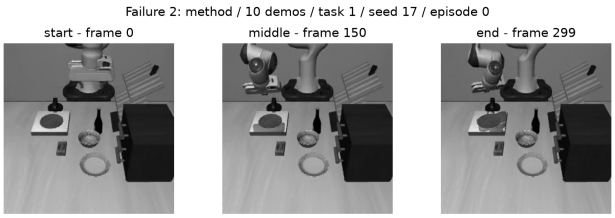
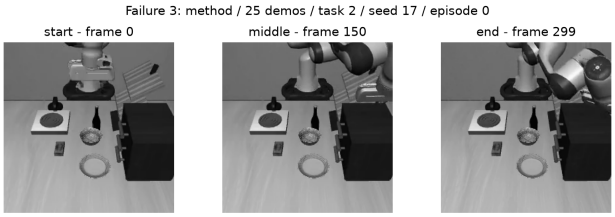
With the correct instruction, mean zero-shot success across task/seed cells is 0.0%; replacing only the instruction with another target task gives 0.0% (drop +0.0 pp). The paired reset seed and environment task are unchanged. Because both conditions are at the performance floor, this control is uninformative about language reliance; it does not establish language invariance.

Task	Seed	Correct prompt	Wrong prompt	Drop
0	17	0.0%	0.0%	+0.0 pp
0	29	0.0%	0.0%	+0.0 pp
1	17	0.0%	0.0%	+0.0 pp
1	29	0.0%	0.0%	+0.0 pp
2	17	0.0%	0.0%	+0.0 pp
2	29	0.0%	0.0%	+0.0 pp

Per-task variation (pooled over two training seeds)

Demos	Task	Naive (n=40)	LoRA+aug (n=40)	Difference
5	0	0.0%	0.0%	+0.0 pp
5	1	0.0%	0.0%	+0.0 pp
5	2	0.0%	0.0%	+0.0 pp
10	0	20.0%	0.0%	-20.0 pp
10	1	0.0%	0.0%	+0.0 pp
10	2	0.0%	0.0%	+0.0 pp
25	0	0.0%	0.0%	+0.0 pp
25	1	0.0%	0.0%	+0.0 pp
25	2	0.0%	0.0%	+0.0 pp

3. Failure analysis

	Failure 1. LoRA+aug, 5 demonstrations, task 0, seed 17, episode 0. Observed. The drawer remains closed while the end effector traverses the central workspace without contact. Hypothesis test - Grounding versus control. Hold state and instruction fixed while changing only target texture/location. Consistent motion toward the wrong object supports grounding failure; correct approach followed by a contact miss supports low-level control failure.
	Failure 2. LoRA+aug, 10 demonstrations, task 1, seed 17, episode 0. Observed. The arm reaches the stove region, but no completed bowl placement is visible at the horizon. Hypothesis test - Action-chunk overshoot. Replay the same initial states with n_action_steps in {1, 5, 10}. Improvement restricted to contact phases under shorter execution supports chunking; an unchanged wrong trajectory does not.
	Failure 3. LoRA+aug, 25 demonstrations, task 2, seed 17, episode 0. Observed. The bottle remains stationary while the arm moves toward the upper-right scene boundary. Hypothesis test - Stage transition. Initialize immediately before and after a subgoal boundary; compare the full instruction with an explicit remaining-step prompt. Improvement only under the latter supports planning/language failure.

Frames are sampled from actual 300-step simulator MP4s. The catalog preserves method, budget, task, training seed, episode id and source video path. These experiments are proposed diagnostic interventions; they are not retroactively treated as performed evidence.

4. Reward-free evaluation and discussion

Bonus B: checkpoint ranking without simulator reward

The small critic is trained from scratch on RGB frames from seen demonstrations using only normalized temporal position; action values are never read. Both critics receive the same deterministic five-rollout sample per checkpoint; Robometer-4B also receives task text. Their scores rank checkpoints and are compared with simulator success using Spearman correlation.

Evaluator	Checkpoints	Spearman rho vs success
Small temporal CNN	42	0.163
Robometer-4B-LIBERO	42	0.279

Direct optimization against either critic is unsafe. The small model can exploit temporal shortcuts and terminal-looking poses; Robometer can be miscalibrated out of distribution. We therefore predefine reward hacking as critic improvement accompanied by lower simulator success on a frozen audit set, and do not mix critic optimization into the main cost curve.

Predictions versus results

We predicted the largest LoRA advantage at five demonstrations. The measured differences were +0.0/-6.7/+0.0 pp at 5/10/25. The result is interpreted from the full per-task/per-seed table rather than only the pooled mean; two seeds remain insufficient for small-effect claims.

Rejected approaches

Approach	Reason for rejection
Full VLM fine-tuning	Poorly identified by 5-25 demos and incompatible with a matched 16 GB budget.
Mixing other held-out tasks	Would improve the headline number through direct target-demo leakage.
Best-demo selection	Adds unreported manual supervision and breaks the fixed-prefix budget.
Open 40-task checkpoint	Its training universe overlaps the held-out goal tasks.

Most unexpected observation

More target data did not help: naive fine-tuning rose to 8/120 at ten demonstrations and returned to 0/120 at 25, while LoRA remained at zero. The result points to update-budget and optimization mismatch rather than simple demo scarcity. Separately, the simulator and data conversion disagree on task ordering, so prompt-based identity checks were indispensable.

Reproducibility

Exact commands, immutable episode prefixes, raw eval_info.json files, 960 outcomes and 480 retained MP4s, checkpoint metadata and aggregation tables are preserved. Sources: SmolVLA (arXiv:2506.01844), LIBERO (arXiv:2306.03310), LeRobot LIBERO documentation and Robometer.